4	**(a)**	p.d. across one lamp = 2.5 V resistance = [(8.7 – 7.5) / 0.3] / 2 = 2.0 Ω	C2 A1
	(b)	straight line through the origin with gradient of 0.5.	M1 A1
	(c)	$P = I^2 R$ $= 0.30^2 \times 2.0$ $= 0.18$ W Alternative method: use $P = V\,I$ or $P = V^2 / R$	 A1
	(d)	**(i)** $R = \rho\, l / A$ $l = (2.0 \times 0.40 \times 10^{-6}) / 1.7 \times 10^{-8}$ $= 47$ m	C1 A1
		(ii) $I = A\,n\,v\,q$ $v = 0.30 / (0.40 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ $= 5.5 \times 10^{-5}$ m s^{-1}	 C1 A1

5	(a)	(i)	force = mg	M1